



W1-2-6-60-1

## JOMO KENYATTA UNIVERSITY OF AGRICULTURE AND TECHNOLOGY

**UNIVERSITY EXAMINATIONS 2025/2026**

SECOND YEAR SEMESTER TWO EXAMINATION FOR THE DEGREE OF MASTERS  
OF SCIENCE IN ARTIFICIAL INTELLIGENCE

**ICS 3308 – DEEP LEARNING**

**Date:** JUNE 2026

**Time:** Practical Exam

**Instructions: Attempt all questions**

### QUESTION ONE

a) You are provided with the following inputs and target output:

| Inputs |       |       |       | Target | Net      | Actual | Error | Weights and Bias |       |       |       |     |
|--------|-------|-------|-------|--------|----------|--------|-------|------------------|-------|-------|-------|-----|
| $X_1$  | $X_2$ | $X_3$ | $X_4$ | $t$    | $y_{in}$ | $y$    | $e$   | $W_1$            | $W_2$ | $W_3$ | $W_4$ | $b$ |
| 1      | 1     | 1     | 1     | 1      |          |        |       |                  |       |       |       |     |
| -1     | 1     | -1    | -1    | 1      |          |        |       |                  |       |       |       |     |
| 1      | 1     | 1     | -1    | -1     |          |        |       |                  |       |       |       |     |
| 1      | -1    | -1    | 1     | -1     |          |        |       |                  |       |       |       |     |

Required:

Design and implement a Perceptron that is able to map all the inputs to their correct target. Initialize all weights and bias to 0 and a learning rate of 1.

(10 Marks)

b) Hopfield networks are one of the earliest RNNs that were developed and used for associative memory. You are required to train a Hopfield network to recall the stored patterns;  $P_1 = [1, -1, 1, -1]$  and  $P_2 = [-1, 1, -1, 1]$ . Show how it behaves given a noisy pattern  $P = [1, 0, 1, -1]$ .

(10 Marks)

**QUESTION TWO**

- a) ABSA bank is using MLP in facilitating credit applications. Using PyTorch-style tensor operations such as `@matmul` or `@add` provide an implementation that implements the forward pass of a 2-layer Multi-Layer Perceptron (MLP) for a single input vector  $X$  (10 Marks)
- b) An intern at Waymo has been requested to demonstrate computer vision data pipelining in an exhibition to a group of investors. The demonstration should:
- Create a TensorFlow pipeline for MNIST dataset that performs Min-Max scaling within a `tf.data` pipeline. (5 Marks)
  - Ensure the data is shuffled, batched, and prefetched before being passed into a custom training loop. (5 Marks)

**QUESTION THREE**

- a) Convolution is a key operation in CNN. Use the input image and the filter given to show the resulting feature map.

|   |   |   |   |   |
|---|---|---|---|---|
| 1 | 1 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 | 0 |
| 0 | 0 | 1 | 1 | 1 |
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 | 0 |

$\otimes$

|   |   |   |
|---|---|---|
| 1 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 1 |

Input image                      Filter

(10 Marks)

- b) Develop a script using either TensorFlow or PyTorch that implements an end-to-end deep learning pipeline. Your solution must satisfy the following technical requirements:
- Load raw data from a NumPy format and implement custom transformation within the pipeline. (3 Marks)
  - Define a Convolutional Neural Network (CNN) architecture designed to process input tensors of shape (32, 32, 3). (4 Marks)
  - Incorporate at least two convolutional layers, utilizing the ReLU activation function to handle non-linearity. (2 Marks)
  - Configure the output to classify the input into 5 distinct categories. (1 Mark)